THE PUNCTURED SPECTRUM OF A NORMAL LOCAL RING IS AN ALGEBRAIC SPACE WITH A GROUPOID PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. The normalization finiteness input is delegated to an uncarded chapter lemma and declared external.

Allowed standard prerequisites (editorial).

- Normalization of the punctured spectrum. lemma-punctured-spec-normalization (same chapter, NOT carded, declared external). The normalization is finite over the punctured spectrum.
- Finite flat equivalence global. Properties of Spaces, proposition finite-flat-equivalence-global (cross-chapter, named). The quotient is an algebraic space.

Source reference. Original reference 02IG: focused-seeds/geometry/S9-decent-punctured-spec/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03IL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2878–2878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BB6: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3489–3489. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 03K6: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2915–2915. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 04ZF: focused-seeds/geometry/S9-decent-punctured-spec/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A4G: focused-seeds/geometry/S9-decent-punctured-spec/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0052: focused-seeds/geometry/S9-decent-punctured-spec/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A31: focused-seeds/geometry/S9-decent-punctured-spec/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0902: focused-seeds/geometry/S9-decent-punctured-spec/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent locally Noetherian algebraic space over S. Let $x \in |X|$. Then*

$$W = \{x' \in |X| : x' \rightsquigarrow x,\ x' \neq x\}$$

is a Noetherian, spectral, sober, Jacobson topological space.

Proof. We may replace by any open subspace containing x. Thus we may assume that X is quasi-compact. Then $|X|$ is a Noetherian topological space (Properties of Spaces, Lemma 04ZF). Thus W is a Noetherian topological space (Topology, Lemma 0052).

Combining Lemma 0BB6 with Properties of Spaces, Lemma 0A4G we see that $|X|$ is a spectral topological space. By Topology, Lemma 0A31 we see that $W \cup \{x\}$ is a spectral topological space. Now W is a quasi-compact open of $W \cup \{x\}$ and hence W is spectral by Topology, Lemma 0902.

Let $E \subset W$ be an irreducible closed subset. Then if $Z \subset |X|$ is the closure of E we see that $x \in Z$. There is a unique generic point $\eta \in Z$ by Proposition 03K6. Of course $\eta \in W$ and hence $\eta \in E$. We conclude that E has a unique generic point, i.e., W is sober.

Let $x' \in W$ be a point such that $\{x'\}$ is locally closed in W. To finish the proof we have to show that x' is a closed point of W. If not, then there exists a nontrivial specialization $x' \rightsquigarrow x'_1$ in W. Let U be an affine scheme, $u \in U$ a point, and let

$U \to X$ be an étale morphism mapping u to x. By Lemma 03IL we can choose specializations $u' \rightsquigarrow u'_1 \rightsquigarrow u$ mapping to $x' \rightsquigarrow x'_1 \rightsquigarrow x$. Let $\mathfrak{p}' \subset \mathcal{O}_{U,u}$ be the prime ideal corresponding to u'. The existence of the specializations implies that $\dim(\mathcal{O}_{U,u}/\mathfrak{p}') \geq 2$. Hence every nonempty open of $\mathrm{Spec}(\mathcal{O}_{U,u}/\mathfrak{p}')$ is infinite by Algebra, Lemma 02IG. By Lemma 03K5 we obtain a continuous map

$$\mathrm{Spec}(\mathcal{O}_{U,u}/\mathfrak{p}') \setminus \{\mathfrak{m}_u/\mathfrak{p}'\} \longrightarrow W$$

Since the generic point of the LHS maps to x' the image is contained in $\overline{\{x'\}}$. We conclude the inverse image of $\{x'\}$ under the displayed arrow is nonempty open hence infinite. However, the fibres of $U \to X$ are finite as X is decent and we conclude that $\{x'\}$ is infinite. This contradiction finishes the proof. $\square$